

Kushal Maktal

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Professional Summary

AI/ML Engineer with 8+ years of experience designing, developing, and deploying machine learning solutions, AI-powered chatbots, and NLP models. Proven expertise in LLMs, Generative AI, recommendation systems, and natural language understanding. Adept at leveraging Google Cloud Services (BigQuery, Looker Studio, Vertex AI, and Dialogflow CX) for seamless data ingestion, model training, and chatbot automation. Strong programming skills in Python, R, and SQL, with a deep understanding of AI/ML algorithms, statistical modeling, and cloud-based deployment. Passionate about solving complex problems through AI-driven automation and conversational AI solutions.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python (NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras), R, SQL, Java, Scala, C++
- **Machine Learning & AI:** Supervised Learning (Regression, Decision Trees, Random Forests, XGBoost, Gradient Boosting), Unsupervised Learning (K-Means, PCA, Hierarchical Clustering), NLP (Text Classification, Named Entity Recognition, Sentiment Analysis, BERT, Transformer models, spaCy, Dialogflow CX, Generative AI, LLMs), Deep Learning (CNNs, RNNs, GANs), Reinforcement Learning
- **Big Data & Data Engineering:** Hadoop, Hive, MapReduce, Apache Kafka, AWS Glue, GCP Dataflow, Redshift, PostgreSQL, SQL Server, Snowflake, BigQuery
- **Cloud Platforms:** AWS (S3, EC2, SageMaker, Redshift, Lambda, Glue), GCP (BigQuery, Vertex AI, Cloud Storage, Dataflow, Google Cloud Services, Dialogflow CX), Azure (Data Factory, Databricks, Synapse Analytics)
- **Visualization & Analytics:** Tableau, Power BI, Looker, Looker Studio, Matplotlib, Seaborn
- **MLOps, DevOps & Deployment:** Docker, Jenkins, Kubernetes, Git, Bitbucket, CI/CD Pipelines, MLflow, Airflow, FastAPI, Auto-scaling, Monitoring Tools (GCP Monitoring, AWS CloudWatch)
- **Advanced Techniques & Explainability:** SHAP, LIME, Autoencoders, Anomaly Detection, Time-Series Forecasting (Prophet, ARIMA), Fraud Detection Models, Computer Vision for Quality Control

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER

RetailMeNot

Austin, TX

JULY 2023 – Present

- Designed and deployed an AI-driven recommendation engine using LLMs and Generative AI, improving user engagement by 20% through personalized deal suggestions.
- Built an advanced chatbot using Dialogflow CX and NLP models, automating customer interactions and reducing support response time by 30%.
- Developed a multi-turn conversational AI workflow leveraging Generative AI and LLMs, enhancing chatbot contextual understanding and dynamic responses.
- Integrated real-time recommendation APIs with chatbot automation, allowing AI-driven deal suggestions based on user preferences and behavior.
- Built scalable data pipelines using Google Cloud Services (BigQuery, Dataflow, Cloud Storage) for training and deploying AI/ML models efficiently.
- Conducted extensive exploratory data analysis (EDA) using BigQuery and Looker Studio to identify customer behavior trends and optimize deal targeting.

- Developed custom NLP models using spaCy, BERT, and Transformers to improve chatbot intent recognition, boosting classification accuracy by 30%.
- Engineered an embedding-based search model using FAISS and vector databases, enabling semantic similarity retrieval for chatbot-generated responses.
- Implemented reinforcement learning techniques to dynamically adjust chatbot responses and deal recommendations, maximizing conversion rates.
- Designed an automated chatbot retraining pipeline with Airflow and CI/CD, reducing deployment time by 40% and ensuring model accuracy.
- Conducted A/B testing to optimize chatbot-generated recommendations, increasing user engagement and improving conversion rates by 15%.
- Created synthetic training datasets using GANs to improve chatbot response diversity and enhance recommendation system generalization.
- Applied Explainable AI techniques using SHAP and LIME, ensuring transparency in chatbot decision-making and AI-driven deal recommendations.
- Integrated a hybrid recommendation system combining collaborative filtering and NLP-based content recommendations, improving deal relevance.
- Built a real-time monitoring dashboard in Looker Studio to track chatbot performance, recommendation success, and customer interaction patterns.
- Implemented weak supervision and active learning techniques to continuously improve chatbot response quality using semi-supervised learning models.
- Developed voice-enabled chatbot capabilities, integrating multi-modal AI to support both text and voice-based interactions for enhanced accessibility.
- Designed a memory-aware chatbot using transformer-based LLMs, enabling personalized and context-aware conversations for returning users.
- Improved chatbot scalability by deploying ML models on Vertex AI and optimizing cloud-based model inference for real-time interactions.
- Enhanced chatbot NLP processing pipeline by fine-tuning BERT-based models for entity recognition and sentiment analysis in customer interactions.
- Developed an AI-driven customer segmentation model using clustering techniques, helping tailor chatbot responses and targeted deal offerings.
- Integrated chatbot automation with enterprise APIs, enabling real-time dynamic responses based on inventory, pricing, and user engagement history.
- Designed a recommendation ranking model using learning-to-rank techniques, improving deal prioritization and increasing conversion efficiency.
- Applied advanced anomaly detection using Autoencoders and GANs to monitor chatbot conversation health and flag potential model drifts.
- Built a multi-agent AI system where chatbot, recommendation engine, and personalization modules communicate for optimized user experience.
- Led cross-functional collaboration with data engineers, software developers, and product teams to align AI-driven solutions with business objectives.

Environment: Python, TensorFlow, BERT, Dialogflow CX, Generative AI, LLMs, Google Cloud Services, BigQuery, Looker Studio, Vertex AI, FAISS, FastAPI, Dataflow, Cloud Storage, Airflow, CI/CD Pipelines, SHAP, LIME, GANs, Auto-scaling, GCP Monitoring Tools.

ML ENGINEER

State of MD

Baltimore, MD

JAN 2022 – MAY 2023

- Designed and deployed an ML-based Medicaid fraud detection system, reducing fraudulent claims and

improving financial oversight.

- Developed a Random Forest and XGBoost model to classify fraudulent vs. non-fraudulent Medicaid claims with improved accuracy.
- Built an anomaly detection model using Autoencoders to flag unusual claims for further investigation.
- Designed data preprocessing pipelines using AWS Glue and Redshift, ensuring scalable processing of large transactional datasets.
- Conducted exploratory data analysis (EDA) and feature engineering to improve fraud classification model performance.
- Integrated real-time fraud alerts into the state's financial monitoring system to assist auditors in identifying suspicious activities.
- Created a Tableau-based analytics dashboard to visualize fraud detection trends and track investigation outcomes.
- Worked with government agencies and compliance teams to ensure machine learning models aligned with regulatory requirements.
- Designed automated model retraining pipelines to adapt to evolving fraud patterns in Medicaid transactions.
- Developed a natural language processing (NLP) module to extract insights from healthcare provider claims and flag inconsistencies.
- Reduced false favorable rates by 18%, improving efficiency for fraud investigation teams.
- Implemented privacy-preserving machine learning techniques to ensure HIPAA and state data regulations compliance.
- Collaborated with data engineers and legal teams to establish the best fraud detection model deployment practices.
- Contributed to policy-making discussions on the adoption of AI-driven fraud prevention techniques.
- Conducted training workshops for government auditors and compliance officers on using machine learning insights for investigations.
- Integrated graph-based machine learning techniques to identify hidden relationships between fraudulent providers and suspicious claims.
- Developed a real-time monitoring system to flag high-risk providers and insurance claims, preventing fraudulent transactions before processing.
- Applied reinforcement learning techniques to optimize audit case prioritization, ensuring that high-value fraud cases are investigated first.
- Designed a self-learning fraud detection model that continuously adapts based on new patterns detected in Medicaid claims.
- Enhanced geospatial analytics integration, helping law enforcement track and predict fraud hotspots based on historical claim locations.

Environment: Python, Scikit-Learn, XGBoost, Autoencoders, AWS Glue, Redshift, Tableau, PostgreSQL, FastAPI, Airflow, SHAP, NLTK, CI/CD Pipelines, Lambda, Bitbucket

DATA/ML ENGINEER

Stryker Corporation

Dallas, TX

OCT 2019 – DEC 2021

- Developed a machine learning-driven quality control system to detect manufacturing defects in medical devices.
- Built classification models (Random Forest, XGBoost) using sensor data, material logs, and production insights to predict defective products.
- Designed an automated data pipeline to process real-time data from IoT-enabled manufacturing equipment.
- Implemented anomaly detection algorithms (Isolation Forests, Autoencoders) to flag deviations in material consistency and machine operation.
- Developed a predictive maintenance model using time-series forecasting to reduce machine downtime.
- Integrated ML models with manufacturing execution systems (MES) to improve production line efficiency.

- Deployed an ML-driven root cause analysis system, identifying patterns leading to defects and optimizing manufacturing parameters.
- Created an interactive dashboard in Power BI to track real-time defect rates and production insights for process engineers.
- Reduced defect rates by 25% by proactively detecting inconsistencies in raw materials and production cycles.
- Lowered production costs by 20% by optimizing manufacturing workflows through ML-driven process automation.
- Conducted feature selection and dimensionality reduction to improve model accuracy while maintaining inference speed.
- Developed and deployed FastAPI microservices to integrate ML-driven quality control insights into existing systems.
- Collaborated with quality assurance teams to implement ML-backed decision support systems for batch testing.
- Ensured compliance with FDA and ISO standards by integrating machine learning techniques for regulatory quality control.
- Provided ML training and mentorship to junior data engineers and quality control specialists to improve manufacturing adoption of AI/ML techniques.
- Designed a computer vision-based defect detection system using Convolutional Neural Networks (CNNs) to enhance product inspections.
- Built a predictive analytics tool that analyzes supplier quality history and predicts potential material failures before manufacturing.
- Developed an automated root cause analysis system that correlates machine sensor anomalies with defect trends to optimize machine configurations.
- Applied unsupervised learning techniques to cluster production failures and identify new potential defect patterns.
- Created a real-time quality feedback loop, allowing ML models to self-adjust parameters based on live factory conditions.

Environment: Python, Scikit-Learn, XGBoost, Random Forest, TensorFlow, AWS Redshift, Snowflake, FastAPI, PostgreSQL, Autoencoders, Docker, Jenkins, Airflow, IoT Sensors, Kubernetes, Power BI

DATA SCIENTIST

Sailpoint

Austin, TX

MAR 2019 – SEP 2019

- Built scalable ETL workflows on AWS for data extraction, preprocessing, and ingestion, improving efficiency by 40%.
- Developed predictive maintenance models using machine learning, reducing equipment downtime by 25%.
- Applied clustering algorithms like K-Means to segment customer behavior, enabling better-targeted marketing campaigns.
- Built APIs for real-time integration of machine learning models into Stryker's data platforms.
- Automated time-series forecasting for supply chain optimization, leading to a 15% reduction in inventory costs.
- Designed Power BI dashboards to provide actionable insights into key operational metrics.
- Conducted feature engineering and model tuning for gradient-boosted machines (GBMs), improving performance by 20%.
- Led the implementation of NLP models for document classification, automating compliance workflows.
- Migrated on-premises data systems to AWS, enabling cost savings and improved scalability.
- Developed anomaly detection systems to identify inefficiencies in manufacturing processes.
- Conducted root cause analysis to resolve data discrepancies in manufacturing datasets.
- Designed data pipelines to preprocess IoT sensor data for machine learning models, improving prediction accuracy by 30%.
- Collaborated with DevOps teams to establish CI/CD pipelines for model deployment, reducing update downtime.
- Implemented explainable AI techniques for compliance, ensuring transparent predictions in regulatory audits.

- Conducted time-series analysis on equipment telemetry data to detect early signs of wear and tear.
- Mentored junior engineers in advanced machine learning techniques and tools.
- Developed a sentiment analysis pipeline for employee surveys, providing actionable insights to HR teams.
- Automated data quality checks to minimize errors in critical workflows.

Environment: Python, TensorFlow, AWS (Glue, Redshift, SageMaker), Power BI, K-Means, Gradient-Boosted Machines (GBMs), FastAPI, Airflow, CI/CD Pipelines, PostgreSQL, Anomaly Detection, Time-Series Forecasting, NLP (Document Classification).

DATA ENGINEER

Cuspyd

Hyderabad, India

JULY 2015 – OCT 2018

- Designed and implemented scalable ETL pipelines for processing data from relational and NoSQL databases, improving data processing efficiency by 30%.
- Migrated legacy systems to the Hadoop ecosystem, enabling the processing of large datasets with high availability and scalability.
- Automated SQL query optimizations for data aggregation workflows, reducing execution times by 45%.
- Developed data pipelines to integrate CRM tools with PostgreSQL, ensuring seamless reporting for marketing and sales teams.
- Built recommendation systems using collaborative filtering techniques to improve product suggestions and drive customer engagement.
- Conducted data validation and implemented error-handling mechanisms to improve pipeline reliability.
- Created real-time dashboards using Power BI to monitor operational metrics, providing actionable insights to business leaders.
- Applied clustering algorithms to segment customer data, enabling targeted marketing campaigns and improving ROI by 20%.
- Developed MapReduce programs to analyze unstructured data and identify trends, enhancing decision-making processes.
- Conducted performance tuning on Hadoop jobs, improving the throughput of batch processing pipelines by 3x.
- Collaborated with cross-functional teams to establish data modeling standards, ensuring consistency across projects.
- Wrote and maintained detailed documentation for ETL workflows, facilitating knowledge transfer and scalability.
- Developed anomaly detection scripts for real-time monitoring of system performance, preventing data processing delays.
- Conducted root cause analysis of data inconsistencies and implemented solutions to ensure data quality and integrity.
- Integrated data from third-party APIs to enhance reporting capabilities, enabling more comprehensive business insights.
- Trained junior team members on Hadoop, Hive, and other big data technologies, fostering a collaborative learning environment.
- Designed scalable data schemas to accommodate the rapid growth of business data requirements.
- Implemented data lake solutions using AWS S3, optimizing storage and retrieval of structured and unstructured data.
- Participated in weekly sprint meetings to prioritize and deliver high-impact data engineering solutions aligned with business objectives.

Environment: Python, Hadoop, Hive, MapReduce, PostgreSQL, Power BI, SQL Server, AWS S3, Data Validation Pipelines, NoSQL (MongoDB, Cassandra), API Integration, Metadata Management.